

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 3 | SLO 1: Thread Pooling and Executors (Code: 3031)

Q: 1. Fill in the blank:

The _____ interface represents an asynchronous execution service in `java.util.concurrent`.

Ans: Answer: `ExecutorService`

Q: 2. True or False:

Creating new threads on demand for every task introduces significant performance overhead.

Ans: Answer: True

Q: 3. Multiple Choice:

Which factory method creates a pool with a set number of threads?

A) `newCachedThreadPool()` B) `newFixedThreadPool()` C) `newSingleThreadExecutor()` D) `newScheduledThreadPool()`

Ans: Answer: B) `newFixedThreadPool()`

Q: 4. Fill in the blank:

Reusing worker threads reduces the overhead of thread _____ and destruction.

Ans: Answer: creation (allocation)

Q: 5. Programming – What happens if `executor.shutdown()` is not called?

Ans: Answer: The JVM may not terminate because active non-daemon pool threads prevent shutdown.

Q: 6. True or False:

`ExecutorService.shutdownNow()` attempts to stop all actively executing tasks.

Ans: Answer: True

Q: 7. Fill in the blank:

The core implementation class behind Executors factory methods is _____.

Ans: Answer: `ThreadPoolExecutor`

Q: 8. Multiple Choice:

Which method is used to submit a task to an `ExecutorService`?

A) `run()` B) `execute()` / `submit()` C) `start()` D) `begin()`

Ans: Answer: B) `execute()` / `submit()`

Q: 9. Fill in the blank:

A pool queue that holds tasks waiting for available threads is typically a _____.

Ans: Answer: `BlockingQueue`

Q: 10. Fill in the blank:

Thread pools prevent resource exhaustion by limiting the maximum number of _____ threads.

Ans: Answer: concurrent (active)